Sriharsha Velicheti

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Summary

Generative AI Engineer with ∼2 years of experience building production-grade AI systems involving LLM orchestration, multimodal RAG pipelines, and agent-based automation. Experienced in designing cloud-native AI architectures integrating retrieval systems, knowledge graphs, and structured tool execution to solve real-world enterprise workflow problems.

Technical Skills

Generative AI: LLM orchestration, Multimodal RAG, AI Agents, Prompt Engineering, Structured Outputs
Programming: Python (FastAPI, Async pipelines), SQL (MySQL), MongoDB, TypeScript
Frameworks: LangChain, Gemini API, OpenAI API, Stagehand, Browserbase, CopilotKit
Cloud & DevOps: GCP (Cloud Run, Vertex AI), Azure AI Services, Docker, GitHub Actions, Azure DevOps
Systems: Microservices architecture, REST APIs, event-driven pipelines, distributed automation workflows

Experience

Data Smith AI — Generative AI Engineer Pune, India

WellsynthAI — Enterprise Oil & Gas Document Intelligence Platform Jan 2026 – Present

- Architected and developed an enterprise AI platform for upstream oil & gas document intelligence, enabling semantic search, grounded chat, automated report generation, and knowledge extraction across End-of-Well Reports, DDRs, drilling programs, and technical documentation.
- Built a production-grade multimodal RAG pipeline using FastAPI, Gemini, Qdrant, PostgreSQL, Redis, and Celery with asynchronous document ingestion, OCR, metadata extraction, vector indexing, and citation-grounded retrieval for large engineering document collections.
- Designed and deployed the complete cloud infrastructure on Azure using Docker, Azure Container Registry, Azure Static Web Apps, Azure VM, and Azure DevOps CI/CD while implementing monitoring, scalable background processing, role-based access, and enterprise-ready API architecture.

AI-Powered GRC Compliance Mapping Engine Jan 2026 – Present

- Built an AI-driven compliance automation system to map security findings to controls across 20+ frameworks (ISO 27001, NIST, SOC2, PCI-DSS) using a multi-stage RAG pipeline (embedding, retrieval, reranking, LLM mapping).
- Designed an LLM-based adversarial validation (critic) module to enforce citation grounding, improve mapping accuracy, and reduce hallucinations for audit-grade outputs.
- Engineered a scalable microservices architecture (FastAPI, Qdrant, Redis, reranker) with LFU caching, achieving low-latency responses and near-zero token cost on repeated queries.

Machine Learning — Physics-Informed Rate of Penetration Prediction 2026

- Developed a physics-informed machine learning framework for predicting drilling Rate of Penetration (ROP) using 19 engineered features, combining drilling parameters, domain-derived ratios, and causal rolling statistics across 7 wells.
- Benchmarked classical Bingham regression and ensemble ML models using well-grouped cross-validation to prevent depth-sequential data leakage, establishing a robust cross-well evaluation framework for ROP prediction.

Blackhat AI Agent — Threat Profile & Attack Scenario Generator Oct 2025 – Present

- Designing an AI-driven adversary simulation platform to generate realistic cyberattack scenarios mapped to the MITRE ATT&CK framework, addressing the need for scalable Breach & Attack Simulation (BAS) environments.
- Built a hybrid reasoning pipeline combining Neo4j knowledge graphs with LLM-driven scenario generation to construct multi-stage adversary campaigns from structured threat intelligence data.
- Integrated MISP threat intelligence feeds to enrich attack profiles with real-world indicators and developed deterministic safety validation rules preventing unsafe or uncontrolled command execution.

Autonomous Court Case Automation — MHADA July 2025 – Sept 2025

- Solved large-scale manual effort in legal case retrieval by developing an automated browser-based data extraction system for multiple dynamic court portals.
- Implemented asynchronous automation pipelines using Python and Playwright to reliably handle dynamic forms, CAPTCHA-aware flows, and structured data capture.
- Delivered measurable operational impact by reducing manual work by 80% and enabling 3× faster case-query processing across distributed court systems.

Go-Bundled Agentic Web Automation Engine Oct 2024 – June 2025

- Built a multi-agent AI automation platform orchestrating task-specific agents for customer support, subscription workflows, and email verification pipelines.
- Implemented browser automation modules using Stagehand + Browserbase enabling reliable execution of signup, update, and cancellation flows across multiple third-party platforms.
- Designed scalable cloud infrastructure with Dockerized services deployed on Google Cloud Run and Cloud Run Jobs, integrating asynchronous execution through Cloud Tasks.

TenderGenie — Document Intelligence Platform Nov 2024 – Oct 2025

- Addressed complex tender-document analysis problems by building a multimodal RAG system capable of understanding PDFs, datasheets, compliance documents, and technical tables.
- Developed structured extraction pipelines using Gemini AI, Qdrant vector search, and FastAPI APIs to support conversational document exploration and summarization.
- Implemented CI/CD pipelines in Azure DevOps and containerized deployments with Docker, enabling automated RAG evaluation and production rollout via Azure App Services.

Projects

Generative UI HR Workspace (TamboChat)

- Built an agentic Generative UI workspace that dynamically renders pre-registered UI components based on user intent, eliminating traditional static menu-driven enterprise navigation.
- Designed a bounded Generative UI architecture combining LLM reasoning with tool schemas using Tambo and Zod to ensure safe and deterministic UI generation.
- Implemented full stack pipeline: AI → Tool → API → Supabase with realtime subscriptions for synchronized workflow state across multiple users.

Multimodal RAG Research System — Siemens Internship

- Researched advanced RAG architectures for complex document ingestion including schema-aware PDF parsing and structured data extraction pipelines.
- Evaluated multiple vector storage strategies and embedding configurations to optimize retrieval accuracy and semantic search relevance.
- Implemented multi-model response evaluation pipelines benchmarking LLM outputs for factual consistency and grounding.

Education

Jain University — Bengaluru, India
Bachelor of Technology in Computer Science (Data Science Honors)
CGPA: 8.756 Aug 2020 – June 2024

Achievements

- TenderGenie product deployed with three enterprise clients within the first year at DataSmithAI.
- Secured 6th position among 159 participants in MACHINE HACK competition with 92.6% model accuracy.
- Conducted Generative AI workshop at I2IT Pune representing DataSmithAI.
- President of Data Science Student Club and mentor to 100+ students in AI/ML learning programs.